

Supplementary Materials for

DrugRepSS: Textual Semantic and Graph Structural

Progressive Representation Learning for Drug Repositioning

1. Supplemental Materials and Methods

1.1 Dataset

The process for the construction of the effectiveness contrast dataset is shown in Figure S1 [1]. Drug effectiveness data are collected and integrated based on the medical literature database PubMed [2] and the medical semantic relational database SemMed [3]. The construction process is divided into two steps:

First, we collected the drug terms from DrugBank [4] and disease terms from Disease Ontology [5], and mapped these terms to the standard terms (i.e. the term headings) in MeSH [6]. We called literature extraction interface in PubMed to extract in batch these terms related 220,425 literatures and their PMIDs, of which the literature types are limited to observational research, RCT research and systematic review research. Based on the literatures obtained above, we extracted a total of 2,460,750 semantic relationships from SemMed. To obtain the drug relationships with effectiveness contrast, we limited the types of entities and their relationships and obtained 137,941 triples among diseases and drugs, which including 35,984 drug-indication relationships. By inviting medical experts to manual review these relationships, 3,314 relationships of drug effectiveness contrast for disease treatment are obtained, including 632 drugs and 264 diseases.

Second, in order to obtain more complete drug effectiveness contrast dataset, we constructed hierarchical relationships of drug effectiveness and extended the dataset (Figure S1B). For example, drug a is more effective than drug b in treating disease c, and drug b that is more effective than drug c in treating disease c, then we can deduce that a is more effective than c. Finally, we totally obtained 3,738 drug-disease

relationships with effectiveness contract information.

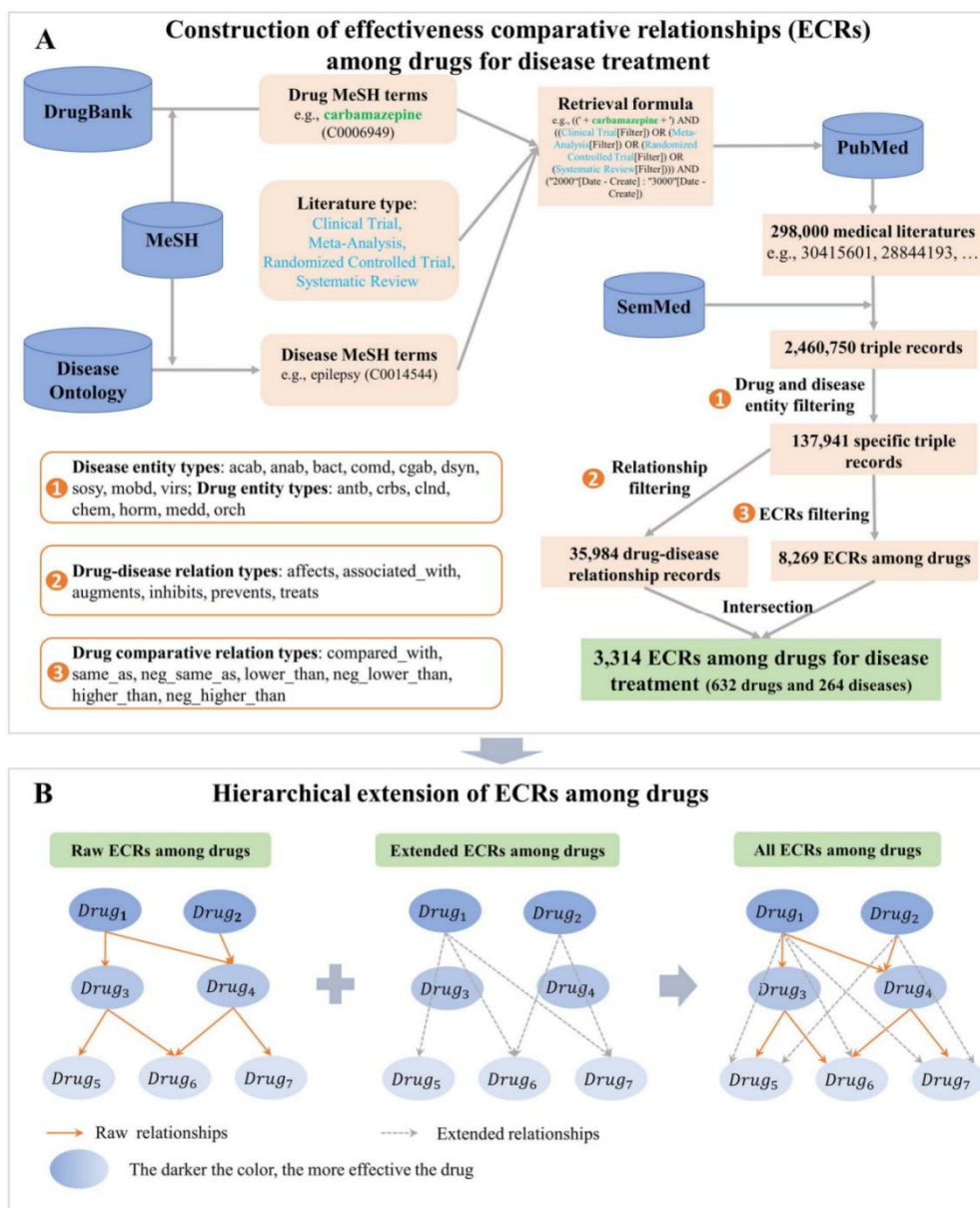


Figure S1. Process for the construction of the drug effectiveness contrast dataset. [1] (A) Construction of drug effectiveness contrast dataset based on literature mining; (B) Hierarchical relationships extension of drug effectiveness contrast.

Table S1. The details of the dataset

Number of Drugs	Number of Diseases	Number of Drug-Disease Associations	Data Consistency
632	264	3738	2.24%

The details of the dataset are presented in Table S1. Some examples of the dataset are given in Table S2 where Disease_CUI and Drug_CUI are IDs from UMLS

database and a higher effectiveness rank indicates better effectiveness.

Table S2. Some examples of the dataset

Disease_name	Disease_CUI	Drug_name	Drug_CUI	Effectiveness rank
Acne	C0702166	Azithromycin	C0052796	1
Acne	C0702166	Clindamycin	C0008947	2
Acne	C0702166	Atenolol	C0004147	3

We divided the 3,738 drug-disease relationships into a training set and a test set at a ratio of 8:2, and then set a ratio of positive and negative samples of 1:5. In generating negative samples, all the relationships between each drug and each disease were first constructed, then the existing drug-disease relationships were removed, and finally a specified number of samples were randomly selected as negative samples from the remaining relationships.

1.2 Baseline Methods for Comparison

In the comparison experiment, we adopted two types of baseline methods. One is the existing drug repositioning methods, and the other is prediction methods based on embedding feature similarity. A detailed description of these baseline methods is described below.

1.5.1 Drug repositioning methods

LAGCN [7] is a drug repositioning model based on Graph Convolutional Network (GCN), which constructs a heterogeneous drug-disease network and applies GCN to predict the drug-disease relationship.

NIMCGCN [8] is a drug repositioning model that combines GCN with a neural induction matrix. It uses GCN from miRNA and disease-like networks to learn the deep features of diseases and employs a new neural induction matrix completion module to predict drug indications.

HNet-DNN [9] is a deep neural network model for drug-disease relationship prediction, which utilizes Deep Neural Network (DNN) to extract deep features from the drug-disease heterogeneous network.

DRONet [1] is a drug repositioning model that combines network embedding and sequencing learning by using the effectiveness comparison relationship between drugs as prior information.

1.5.2 Network embedding methods

1) Matrix decomposition-related methods

SVD is a commonly used matrix decomposition method that can decompose a matrix into the product of three matrices to achieve purposes such as data dimension reduction, denoising, and compression, thereby mapping nodes to a low-dimensional space.

HOPE [10] models the higher-order proximity as a tensor, and then maps the lower-dimensional proximity through tensor decomposition techniques.

GF [11] learns the embeddings of nodes by minimizing the difference between the adjacency matrix and the node representations.

GraRep [12] encodes the network structure information as multiple adjacency matrices of different orders and then maps these adjacency matrices to a low-dimensional space using matrix decomposition techniques.

2) Random walk-related methods

DeepWalk [13] generates node sequences by conducting random walks in the network and utilizes technologies such as Word2Vec to convert these node sequences into low-dimensional vector representations.

Node2vec [14] generates diverse node sequences in the network by controlling the random walk strategy and uses technologies such as Skip-gram to convert the node sequences into low-dimensional vector representations.

3) Neural network-related methods

LINE [15] uses two neural network models to learn the first-order and second-order proximities of nodes and concatenates the learned representations to form the final node embeddings.

SDNE [16] uses a deep autoencoder to learn the embedding representations of nodes, while considering the structural information of the network and the proximities of nodes.

1.3 Evaluation metrics

Hit is one of the significant indicators of ranking learning performance, reflecting the number of relevant drugs included in the query results. *Hit@K* represents the hit rate within the top K results, and the formula is:

$$Hit@k = \frac{1}{n} \sum_{i=1}^n \frac{|s_i|}{|m_i|}$$

where n indicates the number of diseases in the test set, m_i represents the set of drugs corresponding to disease i , and $|m_i|$ represents the total number of drugs in m_i . s_i denotes the intersection of the first K predicted drugs with m_i , and $|s_i|$ represents the number of drugs in the intersection of the first K predicted drugs with m_i .

MRR is a metric for evaluating the performance of LTR on multiple queries. It measures the reciprocal of the average ranking of the highest-ranked drugs among the predicted drugs returned by LTR. For n queries in the test set, $rank_i$ represents the ranking of the highest relevant drug in the ranked list of the i_{th} query disease, and the expression for *MRR* is given in:

$$MRR = \frac{1}{n} \sum_{i=1}^n \frac{1}{rank_i}$$

To more fully assess the experimental results, we plotted the AUROC and AUPRC. Specifically, for each disease, we set a bound q (q is the number of positive case samples for that disease) and set the drug candidates with scores ranked in the top q as having an association with the disease (i.e., label set to 1) and, conversely, as not having an association with the disease (i.e., label set to 0).

1.4 Experimental environment

The experiment in this paper is carried out under the Windows 11 operating system environment and python3.9 language environment, based on CPU. The detailed experimental environment configuration is shown in Table S3.

Table S3. Experimental environment

Setup	Description
CPU	i7-14650Hx
Internal memory	16G
Operating system	Windows 11

2. Supplemental Results

2.1 Comparative experiments of key modules

We present the related experiments and discussions of the key modules, including the LLM-based textual semantic representation module, the structural representation module, and the LTR module.

Table S4. Performance Comparison of DrugRepSS with Different Methods

Types	Methods	HR@1	HR@3	HR@10	MRR
Large Language Models	DrugRepSS _{GPT}	<u>0.1974</u>	0.4087	<u>0.7718</u>	0.3662
	DrugRepSS _{GLM}	0.1718	0.3859	0.7632	0.3437
	DrugRepSS _{LT}	0.1971	0.404	0.7665	0.362
	DrugRepSS _{BG}	0.1977	<u>0.4035</u>	0.7792	<u>0.3656</u>
	DrugRepSS _{BT5}	0.1840	0.3982	0.7672	0.3581
Graph Neural Network	DrugRepSS _{PG}	0.1078	0.2971	0.6890	0.2734
	DrugRepSS _{GCL}	0.1974	0.4087	0.7718	0.3662
	DrugRepSS _{CIN}	<u>0.1685</u>	<u>0.403</u>	<u>0.7483</u>	<u>0.3425</u>
Ranking Learning	DrugRepSS _{RN}	0.1867	0.4003	0.7560	0.3529
	DrugRepSS _{LR}	<u>0.1974</u>	<u>0.4087</u>	<u>0.7718</u>	<u>0.3662</u>
	DrugRepSS _{LM}	0.1571	0.3246	0.6482	0.3047
	DrugRepSS _{PR}	0.2155	0.4259	0.7802	0.3802

DrugRepSS_{GPT}, DrugRepSS_{GLM}, DrugRepSS_{LT}, DrugRepSS_{BG}, and DrugRepSS_{BT5} denote five versions of DrugRepSS based on different LLMs, i.e., GPT, ChatGML, Luotuo, BioGPT, and BioT5; DrugRepSS_{PG}, DrugRepSS_{GCL} and DrugRepSS_{CIN} denote three versions of DrugRepSS based on different GNN algorithms, i.e., All in One, MA_GCL and GIN_Recover; DrugRepSS_{RN}, DrugRepSS_{LR}, DrugRepSS_{LM} and DrugRepSS_{PR} denote four versions of DrugRepSS based on

different LTR algorithms, i.e., RankNet, LambdaRank, LambdaMart and PLRank.

2.1.1 LLM-based textual semantic representation module

We utilize GPT, ChatGLM, LuoTuo, BioGPT, and BioT5 to generate semantic embeddings of drug and disease terms in the dataset, respectively. The experimental results (Table S4) indicate that among these LLM methods, GPT and BioGPT performed the best overall, with similar performance.

2.1.2 Structural representation module

For the structural representation, we adopted three different Graph Neural Network (GNN) methods. They are All in One [17], MA_GCL [18], and GIN_Recover [19]. The experimental results (Table S4) indicate that MA_GCL performs best.

2.1.3 Learning to rank module

For the LTR module of DR, we present comparative results of ranking with four different LTR methods, including RankNet [20], LambdaRank [21], LambdaMart [22], and PLRank [23]. The experimental results (Table S4) show that PLRank performs best, followed by LambdaRank.

2.2 Literature Verification

To verify the potential relationship between diabetes and the predicted other candidate drugs, some possible relationships were found by referring to medical literature in PubMed. For instance, the research by Al-Sofiani et al. indicated that Aspirin (rank=4) helps reduce the risk of cardiovascular disease in patients with diabetes [24]; The study by Gen Yoshino et al. discussed that in hypertensive patients with type 2 diabetes mellitus, nifedipine shows a renoprotective effect similar to ACE inhibitors, which implies that Nifedipine (rank=8) play a certain role in the treatment of diabetes-related complications [25]; The study by S R Davis et al. found that estrogen-progestin (rank=10) therapy may lower the risk of type 2 diabetes [26].

To verify the potential relationship between coronary heart disease and the predicted other candidate drugs, the medical literature in PubMed was referenced. Several possible relationships were found. For example, Schernthaner et al. studied

that Metformin (rank=7) had a beneficial effect on patients with coronary heart disease [27]; The study by Jamal Shamsara et al. showed that pentoxifylline (rank=10) changed protein expression in CAD patients' PBMCs. It can help prevent cardiovascular events and could be beneficial for coronary heart disease [28].

2.3 Network Medicine Analysis

For coronary heart disease, the dense connectivity within the PPI network (34 real links vs. 2.697 expected links, $P=5.66E-26$) suggests that these proteins tend to have a tighter interaction than expected.

2.4 Molecular docking verification

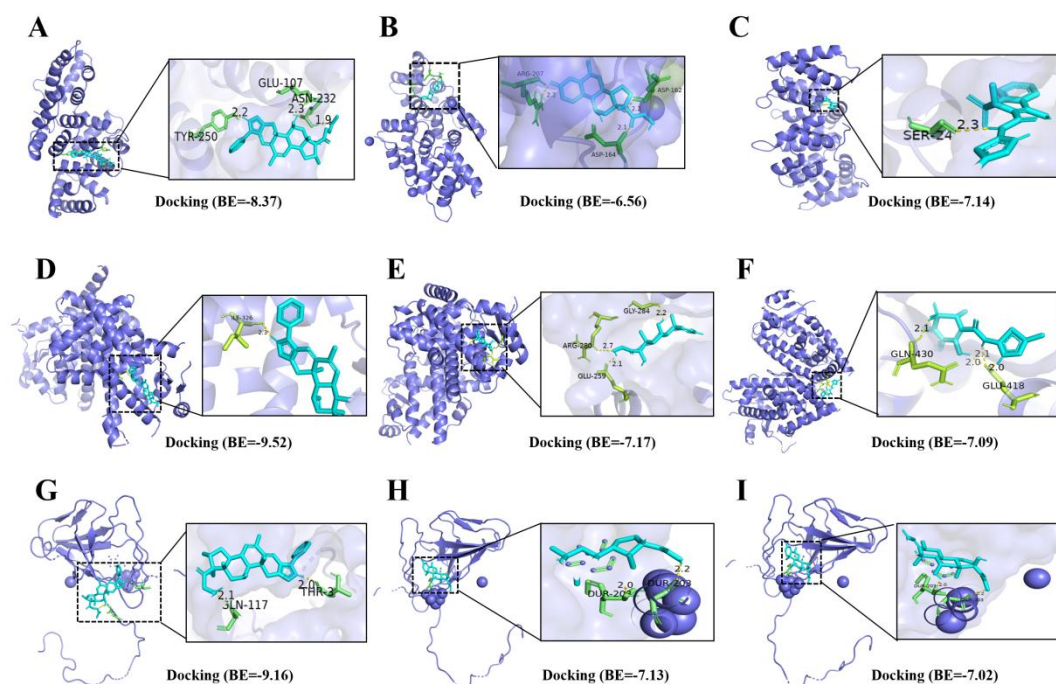


Figure S2. Docking results for diabetes. (A-C) Docking results of the diabetic protein ANXA5 with candidate drugs Adrenal cortex hormones, Methylprednisolone and Meloxicam. (D-F) Docking results of the diabetic protein XDH with candidate drugs Adrenal cortex hormones, Methylprednisolone and Meloxicam. (G-I) Docking results of the diabetic protein XBP1 with candidate drugs Adrenal cortex hormones, Methylprednisolone and Meloxicam.

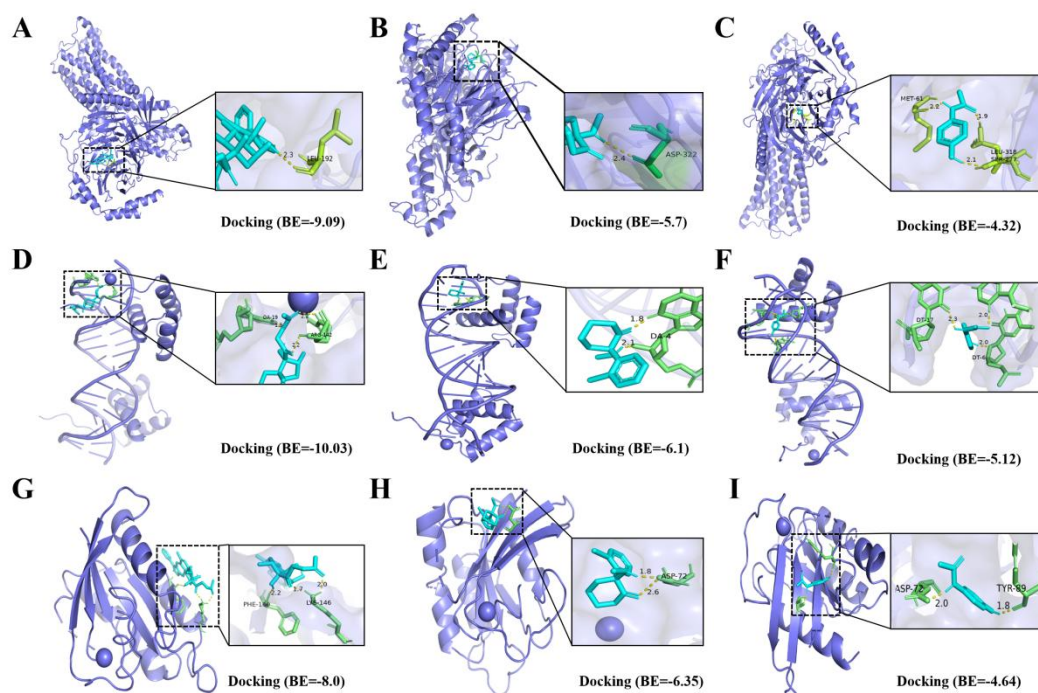


Figure S3. Docking results for coronary heart disease. (A-C) Docking results of the diabetic protein CX3CR1 with candidate drugs Adrenal cortex hormones, Ketamine and Acetaminophen. (D-F) Docking results of the diabetic protein NKX2-5 with candidate drugs Adrenal cortex hormones, Ketamine and Acetaminophen. (G-I) Docking results of the diabetic protein RIT1 with candidate drugs Adrenal cortex hormones, Ketamine and Acetaminophen.

For diabetes, the predicted drugs Adrenal cortex hormones (rank=1), Methylprednisolone (rank=6) and Meloxicam (rank=20) were selected for docking validation. Docking results (Table S5 and Figure S2) show that these drugs has strong binding energies with the diabetes-related proteins ANXA5, XDH and XBP1. And, overall, drugs with higher prediction rankings are more likely to bind to different diabetes-related proteins.

For coronary heart disease, the predicted drugs Adrenal cortex hormones (rank=1), Ketamine (rank=4) and Acetaminophen (rank=15) were selected for docking validation. Docking results (Table S5 and Figure S3) show that Adrenal cortex hormones (rank=1) has strong binding energies ($<-7\text{kcal/mol}$) with the coronary heart disease related proteins CX3CR1, NKX2-5 and RIT1. Ketamine (rank=4) has a certain binding energy ($<-5\text{kcal/mol}$) with CX3CR1, NKX2-5 and RIT1, while

Acetaminophen (rank=15) had a very weak binding energy (>-5 kcal/mol). In general, drugs with higher predicted rankings have stronger binding energies with different diabetes-related proteins.

Table S5. The Binding Energy (kcal/mol) of Various Drugs to Different Proteins Related to Diabetes and Coronary Heart Disease

Disease Name	Protein Name	Drug Name	Ranking of Drugs	Binding Energy(kcal/mol)
Diabetes	ANXA5	Adrenal cortex hormones	1	-8.37
	ANXA5	Methylprednisolone	6	-6.56
	ANXA5	Meloxicam	20	-7.14
	XDH	Adrenal cortex hormones	1	-9.52
	XDH	Methylprednisolone	6	-7.17
	XDH	Meloxicam	20	-7.09
	XBP1	Adrenal cortex hormones	1	-9.16
	XBP1	Methylprednisolone	6	-7.13
	XBP1	Meloxicam	20	-7.02
Coronary Heart Disease	CX3CR1	Adrenal cortex hormones	1	-9.09
	CX3CR1	Ketamine	4	-5.97
	CX3CR1	Acetaminophen	15	-4.32
	NKX2-5	Adrenal cortex hormones	1	-10.03
	NKX2-5	Ketamine	4	-6.1
	NKX2-5	Acetaminophen	15	-5.12
	RIT1	Adrenal cortex hormones	1	-8
	RIT1	Ketamine	4	-6.35
	RIT1	Acetaminophen	15	-4.64

2.5 Analysis of overlapped drugs predicted by DrugRepSS

To verify the specificity of the prediction results, we performed a systematic pairwise analysis of all disease pairs (total $n * (n - 1)/2$ pairs), where n represents the number of diseases. Specifically, for each disease pair, we calculated the overlap of their top 5, 10, 15, and 20 candidate drugs.

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